

DEPARTMENT OF ELECTRICAL & ELECTRONIC ENGINEERING
BANGLADESH UNIVERSITY OF ENGINEERING & TECHNOLOGY
COURSE NO: EEE 202

Name of the Experiment: Study of I-V Characteristics of a BJT

Objective

The objective of this experiment is to study:

- Input characteristics, I_B versus V_{BE}
- Output characteristics, I_C versus V_{CE}
- Parameter extraction (Early voltage V_A and thermal voltage V_T)

Equipment

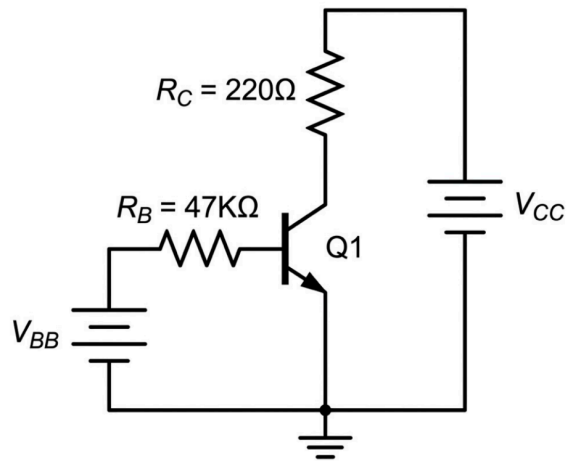
Table 1: List of equipment

Equipment	Specification / Quantity
NPN transistor	2N3904, one piece
Resistors	47 k Ω and 220 Ω
DC source	One unit
Multimeter	One unit
Breadboard	One unit
Jumper wire	A lot

Theory

Circuit Description

The schematic in Fig. 1 depicts an NPN BJT in a common-emitter fixed-bias configuration driven by two DC sources, V_{BB} and V_{CC} .



Common-emitter fixed-bias circuit of an NPN BJT ($R_B = 47 \text{ k}\Omega$, $R_C = 220 \text{ }\Omega$).

Circuit Equations

Applying KVL on the base loop,

$$V_{BB} - I_B R_B - V_{BE} = 0$$

Base current,

$$I_B = \frac{V_{BB} - V_{BE}}{R_B}$$

Collector current,

$$I_C = \beta I_B$$

Emitter current,

$$I_E = \frac{I_C}{\alpha} = (1 + \beta) I_B$$

Common-emitter current gain (with $\beta \gg 1$),

$$\beta = \frac{I_C}{I_B}$$

Applying KVL on the collector loop,

$$V_{CC} - I_C R_C - V_{CE} = 0$$

Collector-emitter voltage,

$$V_{CE} = V_{CC} - I_C R_C$$

Governing equations,

$$I_B = \frac{V_{BB} - V_{BE}}{R_B}$$

$$I_C = \frac{V_{CC} - V_{CE}}{R_C}$$

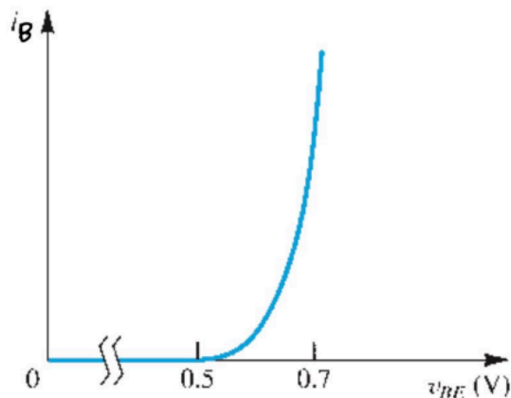
Modes of Operation

Table 2: BJT modes of operation (NPN transistor)

Mode	Emitter-Base Junction Bias	Collector-Base Junction Bias	V_{BE}	V_{CB}	V_{CE}
Active	Forward	Reverse	$\approx 0.7 \text{ V}$	$> -0.4 \text{ V}$	$> 0.3 \text{ V}$
Saturation	Forward	Forward	$\approx 0.7 \text{ V}$	$< -0.4 \text{ V}$	$\approx 0.2 \text{ V}$
Cutoff	Reverse	Reverse	$< 0.7 \text{ V}$	$> -0.4 \text{ V}$	$= V_{CC}$
Inverse active	Reverse	Forward	$< 0.7 \text{ V}$	$< -0.4 \text{ V}$	< 0

Input Characteristics

The input characteristic is the plot of the base current I_B against the base-emitter voltage V_{BE} . The base current is negligible until V_{BE} reaches the cut-in voltage of about 0.5 V , and then rises exponentially as the base-emitter junction becomes forward biased (Fig. 2).



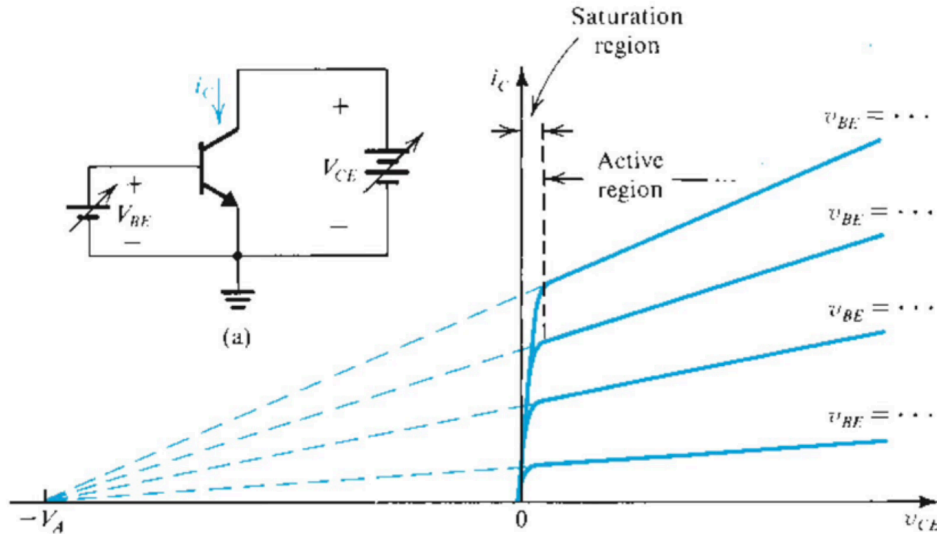
The i_B - v_{BE} characteristic for an NPN transistor:

$$I_B = I_S e^{V_{BE}/V_T}$$

where I_S is the saturation (scale) current and V_T is the thermal voltage.

Output Characteristics

The output characteristic is the plot of the collector current I_C against the collector-emitter voltage V_{CE} for a constant base current (or constant V_{BE}). It has a saturation region at small V_{CE} and an active region in which I_C rises slightly with V_{CE} because of base-width modulation. Extrapolating the active-region lines backward, they meet the V_{CE} axis at $-V_A$, where V_A is the Early voltage (Fig. 3).



The i_C - v_{CE} characteristics of a practical BJT: (a) measurement circuit, (b) characteristic curves.

In the active region,

$$I_C = I_S e^{V_{BE}/V_T} \left(1 + \frac{V_{CE}}{V_A} \right)$$

In active mode, the current gain is

$$\beta = \frac{I_C}{I_B}$$

Procedure

Part A: Input Characteristics (I_B vs. V_{BE} at Constant V_{CE})

- (1) **Initial circuit setup.** Construct the common-emitter fixed-bias circuit on the breadboard with $R_B = 47 \text{ k}\Omega$ at the base terminal and $R_C \approx 220 \text{ }\Omega$ at the collector terminal. Ensure both variable DC power supplies (V_{BB} and V_{CC}) are initially set to 0 V before powering on the circuit.

- (2) **Instrument connection.** Connect a voltmeter directly across the base-emitter terminals to measure V_{BE} and a second voltmeter across the collector-emitter terminals to monitor V_{CE} . Connect a microammeter in series with the base branch (or measure the voltage drop across R_B) to determine the base current I_B .
- (3) **Setting the constant output parameter (V_{CE}).** Adjust the collector supply V_{CC} while keeping V_{BB} at 0.2 V so that the collector-emitter voltage V_{CE} is established at a constant value of 5 V.
- (4) **Stepping the input supply (V_{BB}).** Gradually increase the base supply voltage V_{BB} in small increments through the cut-in region and then in larger steps once the base-emitter junction is fully forward biased.
- (5) **Compensating for the collector voltage drop (crucial step).** As V_{BB} increases, the base current I_B rises, causing the collector current $I_C = \beta I_B$ and the voltage drop across R_C ($I_C R_C$) to increase. Because

$$V_{CE} = V_{CC} - I_C R_C$$

V_{CE} will drop below 5 V if V_{CC} is left untouched. At every single step of V_{BB} , manually increase V_{CC} until the voltmeter across the collector-emitter terminals reads exactly $V_{CE} = 5.0$ V.

- (6) **Data recording.** Once $V_{CE} = 5.0$ V is locked in for a given V_{BB} setting, record the corresponding readings of V_{BE} (in V), I_B (in μ A), and the required V_{CC} (in V) in Table 3.

Table 3: Data for input characteristics ($V_{CE} = 5.0$ V)

Sl.	V_{BB} (V)	V_{BE} (V)	I_B (μ A)	V_{CC} (V)	V_{CE} (V)
1					5.0
2					5.0
3					5.0
4					5.0
5					5.0
6					5.0
7					5.0
8					5.0

Sl.	V_{BB} (V)	V_{BE} (V)	I_B (μ A)	V_{CC} (V)	V_{CE} (V)
9					5.0
10					5.0

Part B: Output Characteristics (I_C vs. V_{CE} at Constant I_B)

- (1) **Establishing a fixed base current (I_B).** Reset V_{CC} to 0 V. Adjust the base power supply V_{BB} to 3.0 V so that the voltage drop across the 47 k Ω base resistor is approximately 2.3 V, establishing a constant base current of

$$I_B = \frac{V_{BB} - V_{BE}}{R_B} \approx \frac{2.3 \text{ V}}{47 \text{ k}\Omega} \approx 49 \mu\text{A}$$

- (2) **Instrument connection for the output loop.** Keep the voltmeter across the collector-emitter terminals to measure V_{CE} , and connect a milliammeter in series with the collector branch (or measure the voltage drop across R_C) to obtain I_C .
- (3) **Sweeping the collector supply (V_{CC}).** Increase the collector supply voltage V_{CC} from 1.0 V to 7.0 V in uniform steps of 0.5 V.
- (4) **Maintaining constant I_B across regions.** As the BJT transitions from the saturation region ($V_{CE} < 0.3 \text{ V}$) into the forward-active region ($V_{CE} > 0.3 \text{ V}$), base-width modulation slightly alters V_{BE} . Fine-tune V_{BB} at each step if needed so that I_B remains strictly constant.
- (5) **Data recording.** For each 0.5 V increment of V_{CC} , record the corresponding collector-emitter voltage V_{CE} (in V) and collector current I_C (in mA) in Table 4.

Table 4: Data for output characteristics (constant $I_B = \underline{\hspace{1cm}} \mu\text{A}$)

Sl.	V_{CC} (V)	V_{CE} (V)	I_C (mA)
1	1.0		
2	1.5		
3	2.0		
4	2.5		
5	3.0		
6	3.5		
7	4.0		

Sl.	V_{CC} (V)	V_{CE} (V)	I_C (mA)
8	4.5		
9	5.0		
10	5.5		
11	6.0		
12	6.5		
13	7.0		

Report

- (1) Plot the input characteristic curve (I_B vs. V_{BE}) at a constant collector-emitter voltage (V_{CE}) using the experimental data.
- (2) Plot the output characteristic curve (I_C vs. V_{CE}) at a constant base current (I_B) using the experimental data.
- (3) Determine the Early voltage (V_A) of the BJT from the output characteristic curve.
- (4) Determine the threshold/thermal voltage (V_T) of the BJT from the input characteristic curve.